

Redox project

Submitted by : **Gokul Udayakumar**

06-09-2023

Modified on : **21-09-2023**

1. Dealing with the noisy data

The data was behaving very strangely, hence a simple 3 sigma outlier cut along with a 10 segment rolling mean is done in order to remove the noise from the data.

The python code used for this step is as follows:

```
#####
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import os

# List of sites and heights
sites = ['LM', 'HMG', 'PZ', 'HM']
heights = ['5', '10', '30', '60']

# Loop over each site and height
for site in sites:
    for height in heights:
        # Defining the filename based on site and height
        filename = f'{site}_{height}.csv'
        print(f"Processing {filename}...")

        data = pd.read_csv(filename, parse_dates=['TimeStamp'])
        data = data.dropna()

        # Subplots
        fig, axes = plt.subplots(2, 1, figsize=(10, 10), sharex=True)

        # Original Data Plot (on the top)
        axes[0].set_title(f'Original Data - {site} {height} cm')
        for column in data.columns[1:]:
```

```

        axes[0].plot(data['TimeStamp'],
data[column],label=column)
        axes[0].set_ylabel('Value')
        axes[0].legend()
        axes[0].grid()

        # Apply 3-std filter and rolling mean
        for column in data.columns[1:]:
            mean = data[column].mean()
            std = data[column].std()
            threshold = 3 * std
            data[column] = np.where(np.abs(data[column] - mean) >
threshold, np.nan, data[column])
            data[column] = data[column].rolling(window=10,
min_periods=1).mean()

        # Filtered Data Plot (to the bottom)
        axes[1].set_title(f'Filtered Data (3-std + 10-segment Mean) -
{site} {height} cm')
        for column in data.columns[1:]:
            axes[1].plot(data['TimeStamp'], data[column],
label=column)
            axes[1].set_xlabel('Time')
            axes[1].set_ylabel('Value')
            axes[1].legend()
            axes[1].grid()

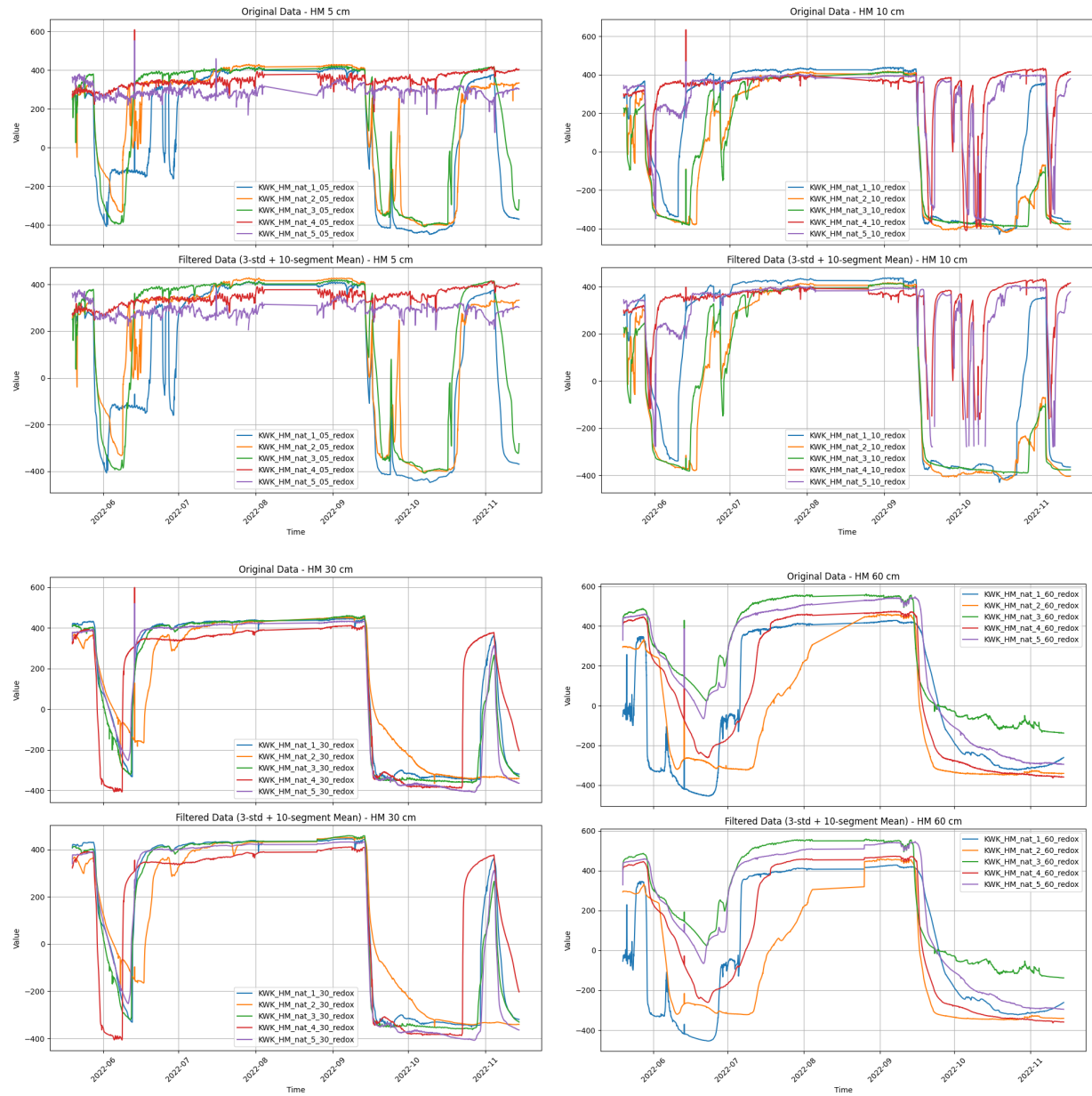
        #axes[0].set_ylim(-2000, 2500)
        #axes[1].set_ylim(-2000, 2500)
        output_folder = 'Filtered'
        os.makedirs(output_folder, exist_ok=True)
        data.to_csv(os.path.join(output_folder,
f'{site}_{height}_modified.csv'))
        plt.xticks(rotation=45)
        plt.tight_layout()
        plt.savefig(os.path.join(output_folder,
f'{site}_{height}.png'), dpi=100)
        plt.close()

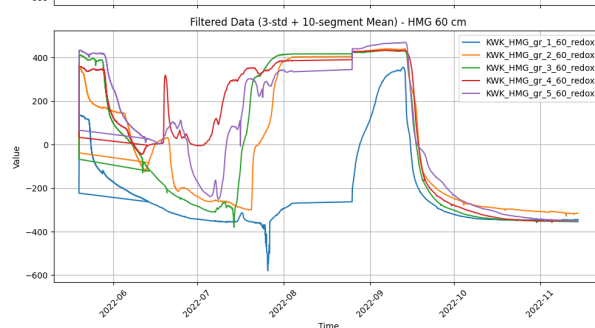
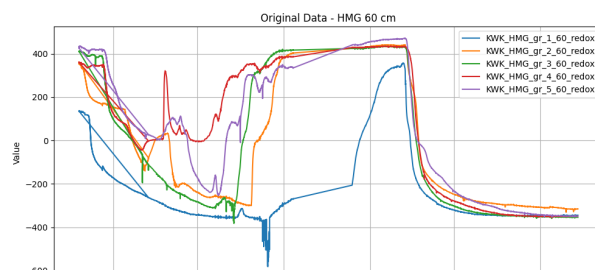
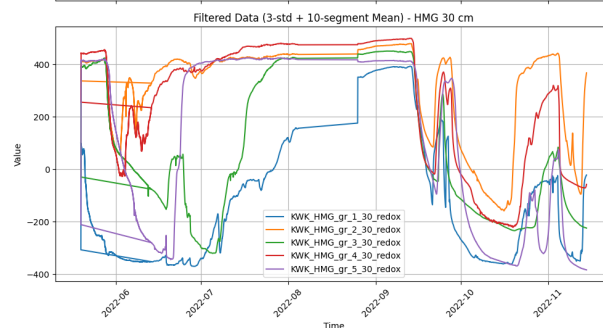
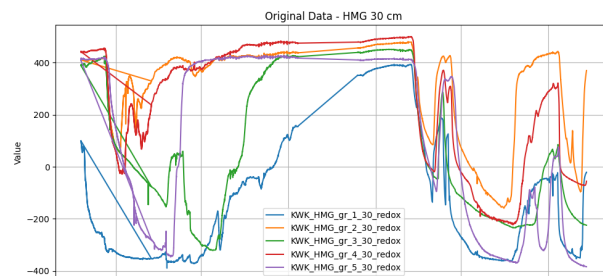
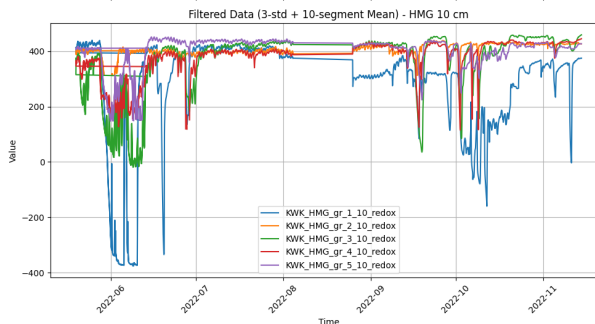
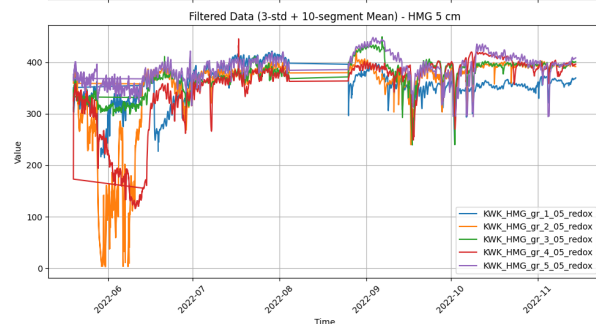
print("done.")

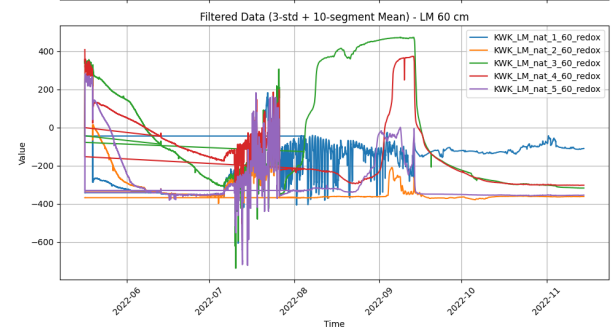
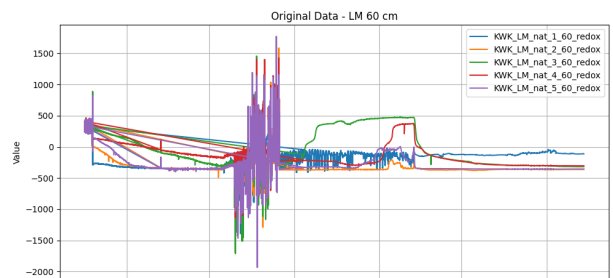
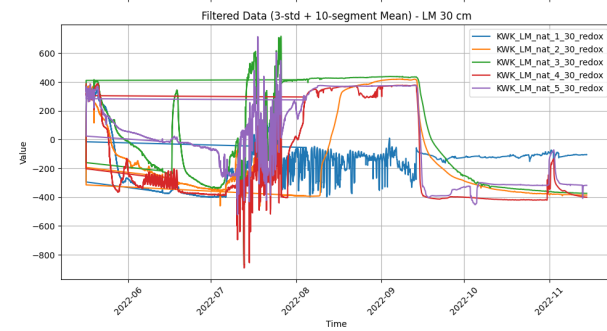
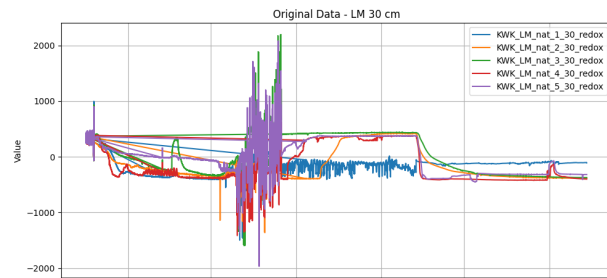
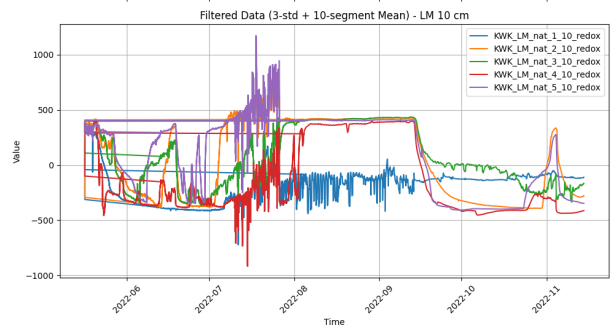
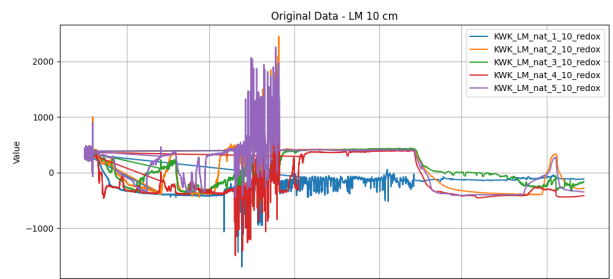
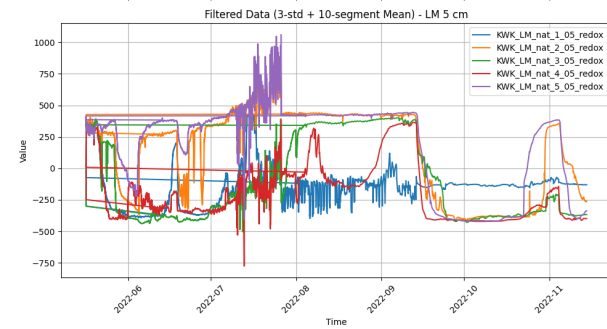
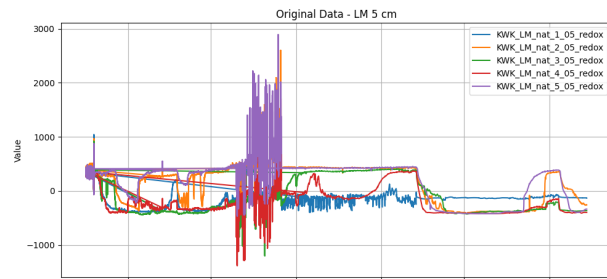
```

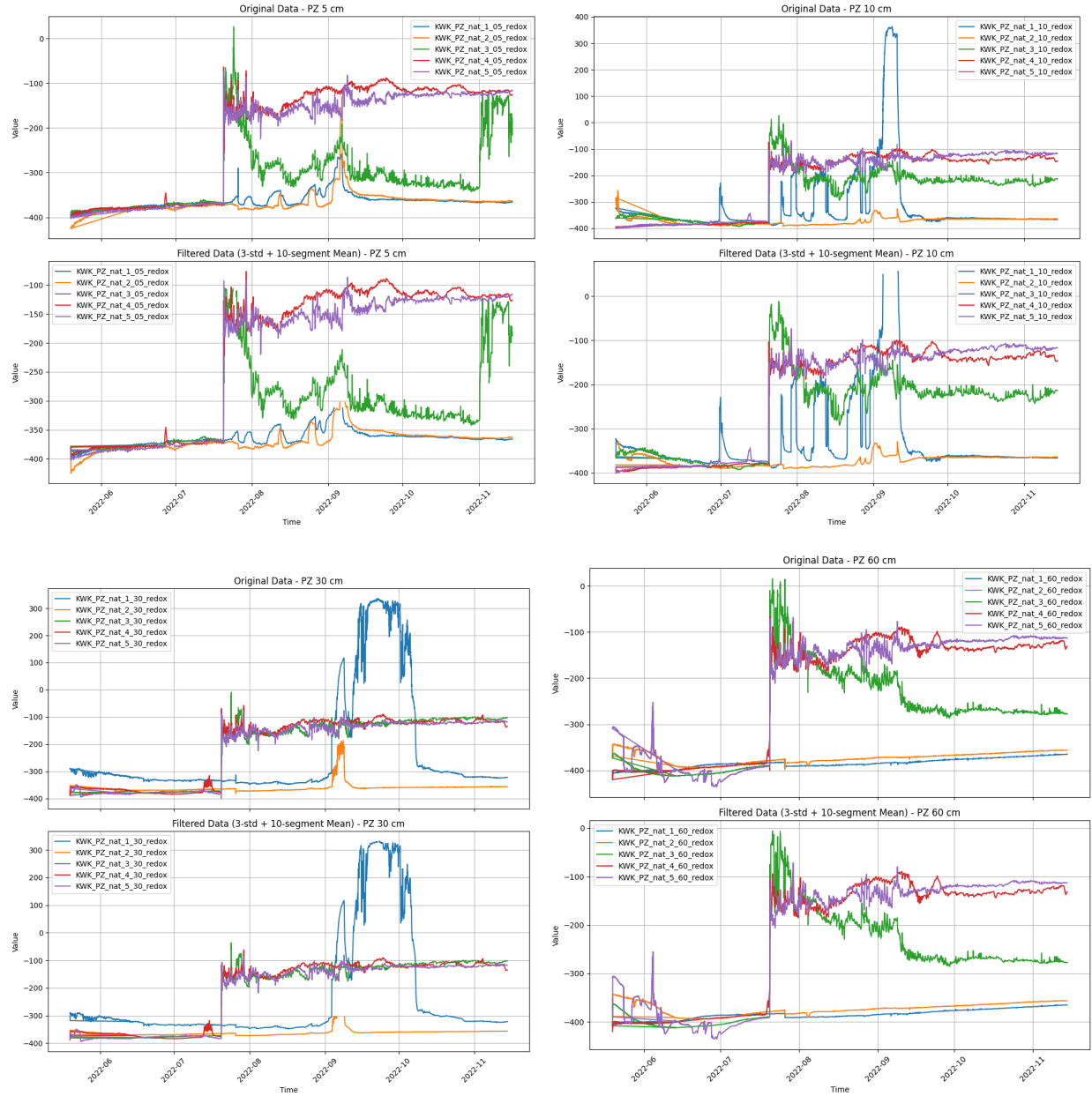
```
#####
```

The figures below display both the initial raw data and the filtered data for reference purposes. In each figure, the upper plot illustrates the raw data, while the lower plot depicts the data after noise removal and the removal of outliers. Each plot corresponds to a specific height level from each measurement site.









2. Finding the parallel

Even after applying noise cancellation, there is notable disparity among the readings from all the sensors during the same timeframe. This discrepancy makes it challenging to identify sensors with comparable measurements. To address this issue, various statistical methods are employed to identify similarities.

Given that the measurements do not exhibit consistent patterns with respect to seasonality or an overarching timescale, a comprehensive analysis of the entire dataset would not yield satisfactory comparisons. Consequently, the dataset is segmented into two distinct time periods: **summer (from July 15, 2022, to September 10, 2022)** and **winter (October 2022)**. It's worth noting that these timeframes

were chosen based on the characteristics of the datasets rather than adhering to the traditional definition of seasons in Germany.

1. First method :- One way ANOVA

A one-way ANOVA is a parametric statistical test that assumes the datasets being compared follow a normal distribution. In this analysis, comparisons are conducted independently at each depth level across all measurement sites. The datasets used for the analysis have been modified after applying noise cancellation techniques. The primary objective of this test is to determine whether all the measurements, provided by the five sensors within each height-site group, exhibit similar distributions. When the p-value in a statistical test falls below the chosen significance level (α , typically set at 0.05), it signifies statistical significance.

Null Hypothesis (H₀): The null hypothesis states that there are no significant differences among the group means. In other words, it assumes that all the group means are equal.

Alternative Hypothesis (H_a): The alternative hypothesis, on the other hand, asserts that there is at least one significant difference among the group means. It suggests that at least one group-mean is different from the others.

The code used for the analysis is as follows:-

```
** This particular code is for the winter period only, For summer, the dates has to be modified accordingly **
```

```
#####
```

```
import pandas as pd
import scipy.stats as stats

dataset_files = [
    'HM_10_modified.csv', 'HM_30_modified.csv', 'HM_5_modified.csv',
    'HM_60_modified.csv',
    'HMG_10_modified.csv', 'HMG_30_modified.csv',
    'HMG_5_modified.csv', 'HMG_60_modified.csv',
    'LM_10_modified.csv', 'LM_30_modified.csv', 'LM_5_modified.csv',
    'LM_60_modified.csv',
    'PZ_10_modified.csv', 'PZ_30_modified.csv', 'PZ_5_modified.csv',
    'PZ_60_modified.csv'
]

results = []
```

```

# Looping through the dataset files
for dataset_file in dataset_files:
    data = pd.read_csv(dataset_file)
    data['TimeStamp'] = pd.to_datetime(data['TimeStamp'])
    data = data.drop(columns=['Unnamed: 0'])

    # Extracting the winter period data using only october
    start_date = '2022-10-01'
    end_date = '2022-10-31'
    summer_data = data[(data['TimeStamp'] >= start_date) &
    (data['TimeStamp'] <= end_date)]
    summer_data.drop(columns='TimeStamp', inplace=True)

    # Checking for missing values
    if summer_data.isnull().values.any():
        print(f"Warning: Missing values found in {dataset_file}.
Handling missing values...")
        summer_data.fillna(summer_data.mean(), inplace=True)

    # one-way ANOVA test
    result = stats.f_oneway(*summer_data.values)
    p_value = result.pvalue

    # Check if the null hypothesis is rejected
    null_hypothesis_rejected = p_value < 0.05

    # Append the results to the list
    results.append({'Dataset': dataset_file, 'P-Value': p_value, 'Null
Hypothesis Rejected': null_hypothesis_rejected})

# Create a DataFrame from the results list
results_df = pd.DataFrame(results)

# Print the results
print(results_df)

#####

```

Results are appended below in the form of a table for convenience, The sites where the null hypothesis is rejected are highlighted with green colour.

for summer:

Dataset	P value	Null hypothesis rejected
HM_05_modified.csv	1	FALSE
HM_10_modified.csv	1	FALSE
HM_30_modified.csv	0.6690490414	FALSE
HM_60_modified.csv	1	FALSE
HMG_05_modified.csv	2.11E-50	TRUE
HMG_10_modified.csv	1	FALSE
HMG_30_modified.csv	1	FALSE
HMG_60_modified.csv	1.75E-195	TRUE
LM_05_modified.csv	1	FALSE
LM_10_modified.csv	1	FALSE
LM_30_modified.csv	1.11E-234	TRUE
LM_60_modified.csv	0.006524362373	TRUE
PZ_05_modified.csv	1	FALSE
PZ_10_modified.csv	2.13E-06	TRUE
PZ_30_modified.csv	0.0632903831	FALSE
PZ_60_modified.csv	1	FALSE

for Winter:

Dataset	P value	Null hypothesis rejected
HM_05_modified.csv	2.11E-183	TRUE
HM_10_modified.csv	1	FALSE
HM_30_modified.csv	3.18E-05	TRUE

HM_60_modified.csv	1	FALSE
HMG_05_modified.csv	8.40E-248	TRUE
HMG_10_modified.csv	0.7140946581	FALSE
HMG_30_modified.csv	2.37E-133	TRUE
HMG_60_modified.csv	0	TRUE
LM_05_modified.csv	1.02E-29	TRUE
LM_10_modified.csv	1	FALSE
LM_30_modified.csv	1	FALSE
LM_60_modified.csv	1	FALSE
PZ_05_modified.csv	1	FALSE
PZ_10_modified.csv	1	FALSE
PZ_30_modified.csv	1	FALSE
PZ_60_modified.csv	1	FALSE

2. Second method : Kruskal wallis test

The Kruskal-Wallis test is a non-parametric statistical test used to determine if there are significant differences between three or more independent groups. It is often used as an alternative to the one-way ANOVA when the assumptions of normality and homogeneity of variances are not met. In the Kruskal-Wallis test, the null hypothesis (H0) and alternative hypothesis (Ha) can be stated as follows:

Null Hypothesis (H0): The null hypothesis in the Kruskal-Wallis test states that there are no significant differences among the medians of the groups being compared. It assumes that all groups come from the same population or have the same median.

Alternative Hypothesis (Ha): The alternative hypothesis, on the other hand, asserts that there is at least one group that is different from the others in terms of median values. In other words, it suggests that there are significant differences among the groups.

*** This particular code is for the winter period only, For summer, the dates has to be modified accordingly ***

```
#####
```

```
import pandas as pd
import scipy.stats as stats

dataset_files = [
    'HM_10_modified.csv', 'HM_30_modified.csv', 'HM_5_modified.csv',
    'HM_60_modified.csv',
    'HMG_10_modified.csv', 'HMG_30_modified.csv',
    'HMG_5_modified.csv', 'HMG_60_modified.csv',
    'LM_10_modified.csv', 'LM_30_modified.csv', 'LM_5_modified.csv',
    'LM_60_modified.csv',
    'PZ_10_modified.csv', 'PZ_30_modified.csv', 'PZ_5_modified.csv',
    'PZ_60_modified.csv'
]

results = []

# Loop through the dataset files
for dataset_file in dataset_files:
    # Read the dataset
    data = pd.read_csv(dataset_file)
    data['TimeStamp'] = pd.to_datetime(data['TimeStamp'])
    data = data.drop(columns=['Unnamed: 0'])

    # Extract the winter period data (October)
    start_date = '2022-10-01'
    end_date = '2022-10-31'
    summer_data = data[(data['TimeStamp'] >= start_date) &
        (data['TimeStamp'] <= end_date)]
    summer_data.drop(columns='TimeStamp', inplace=True)

    # Check for missing values and handle them
    if summer_data.isnull().values.any():
        print(f"Warning: Missing values found in {dataset_file}.
Handling missing values...")
        summer_data.fillna(summer_data.mean(), inplace=True)

    # Perform Kruskal-Wallis test
    result = stats.kruskal(*summer_data.values, nan_policy='omit')

    # Get the p-value
    p_value = result.pvalue
```

```

# Check if the null hypothesis is rejected
null_hypothesis_rejected = p_value < 0.05

# Append the results to the list
results.append({'Dataset': dataset_file, 'P-Value': p_value,
'Null Hypothesis Rejected': null_hypothesis_rejected})

# Create a DataFrame from the results list
results_df = pd.DataFrame(results)

# Print the results
print(results_df)

```

#####

Summer KWT

Dataset	P value	Null hypothesis rejected
HM_05_modified.csv	1	FALSE
HM_10_modified.csv	1	FALSE
HM_30_modified.csv	1.25E-36	TRUE
HM_60_modified.csv	1	FALSE
HMG_05_modified.csv	1.45E-11	TRUE
HMG_10_modified.csv	1	FALSE
HMG_30_modified.csv	1	FALSE
HMG_60_modified.csv	0	TRUE
LM_05_modified.csv	1	FALSE
LM_10_modified.csv	1	FALSE
LM_30_modified.csv	2.20E-174	TRUE
LM_60_modified.csv	1.46E-202	TRUE
PZ_05_modified.csv	1	FALSE

PZ_10_modified.csv	0.9607018882	FALSE
PZ_30_modified.csv	9.95E-132	TRUE
PZ_60_modified.csv	1	FALSE

Winter KWT

Dataset	P value	Null hypothesis rejected
HM_05_modified.csv	1.97E-127	TRUE
HM_10_modified.csv	1	FALSE
HM_30_modified.csv	1	FALSE
HM_60_modified.csv	1	FALSE
HMG_05_modified.csv	1.73E-49	TRUE
HMG_10_modified.csv	1.46E-73	TRUE
HMG_30_modified.csv	1.95E-98	TRUE
HMG_60_modified.csv	3.51E-295	TRUE
LM_05_modified.csv	7.85E-196	TRUE
LM_10_modified.csv	1	FALSE
LM_30_modified.csv	1	FALSE
LM_60_modified.csv	1.46E-202	TRUE
PZ_05_modified.csv	1	FALSE
PZ_10_modified.csv	1	FALSE
PZ_30_modified.csv	1	FALSE
PZ_60_modified.csv	1	FALSE


```
summer_data.drop(columns='TimeStamp', inplace=True)

# Perform Dunn's post hoc test
results = sp.posthoc_dunn([list(summer_data.iloc[:,i]) for i in
range(len(summer_data.columns))])
results.columns=summer_data.columns
results.index=summer_data.columns
results
# Create a table to indicate significant pairs (the p value threshold
is taken down to 0.01 from 0.05 for convenience)
significant_pairs = (results > 0.01)
results
significant_pairs ###(remove the command inorder to see the table as
true or false instead of p values)

#####
```

Results for Summer

HM_5

In the context of post hoc tests, when the null hypothesis is that there are no statistical differences between the groups, a p-value less than 0.01 typically means that you should reject the null hypothesis. This rejection implies that the groups are not statistically similar, and there are significant differences between them. Therefore, you may consider placing them into separate categories or treating them as distinct groups. The results for the comparison of sensors at the depth of 5 cm at site High Marsh shows the p values as follows. This table is obtained from the above code.

	KWK_HM_nat_1_05_redox	KWK_HM_nat_2_05_redox	KWK_HM_nat_3_05_redox	KWK_HM_nat_4_05_redox	KWK_HM_nat_5_05_redox
KWK_HM_nat_1_05_redox	1.000000e+00	0.000000e+00	1.459804e-91	0.000000e+00	0.000000e+00
KWK_HM_nat_2_05_redox	0.000000e+00	1.000000e+00	2.141453e-186	0.000000e+00	0.000000e+00
KWK_HM_nat_3_05_redox	1.459804e-91	2.141453e-186	1.000000e+00	0.000000e+00	0.000000e+00
KWK_HM_nat_4_05_redox	0.000000e+00	0.000000e+00	0.000000e+00	1.000000e+00	7.222461e-238
KWK_HM_nat_5_05_redox	0.000000e+00	0.000000e+00	0.000000e+00	7.222461e-238	1.000000e+00

For convenience, the tagged part (## significant pairs) is activated and the results are converted into True or False values. TRUE means the pairs are statistically significant. And FALSE means they are different. The aim here is to look for pairs with significant relation or can be called parallels.

	KWK_HM_nat_1_05_redox	KWK_HM_nat_2_05_redox	KWK_HM_nat_3_05_redox	KWK_HM_nat_4_05_redox	KWK_HM_nat_5_05_redox
KWK_HM_nat_1_05_redox	True	False	False	False	False
KWK_HM_nat_2_05_redox	False	True	False	False	False
KWK_HM_nat_3_05_redox	False	False	True	False	False
KWK_HM_nat_4_05_redox	False	False	False	True	False
KWK_HM_nat_5_05_redox	False	False	False	False	True

No significant pairs are found for HM_5

Further results are shown for only the selected pairs. Pairs are selected based on Kruskal wallis test results.

HM_30

	KWK_HM_nat_1_30_redox	KWK_HM_nat_2_30_redox	KWK_HM_nat_3_30_redox	KWK_HM_nat_4_30_redox	KWK_HM_nat_5_30_redox
KWK_HM_nat_1_30_redox	True	False	False	False	False
KWK_HM_nat_2_30_redox	False	True	False	False	False
KWK_HM_nat_3_30_redox	False	False	True	False	False
KWK_HM_nat_4_30_redox	False	False	False	True	False
KWK_HM_nat_5_30_redox	False	False	False	False	True

No significant pairs are found for HM_30

HMG_5

	KWK_HMG_gr_1_05_redox	KWK_HMG_gr_2_05_redox	KWK_HMG_gr_3_05_redox	KWK_HMG_gr_4_05_redox	KWK_HMG_gr_5_05_redox
KWK_HMG_gr_1_05_redox	True	False	False	False	False
KWK_HMG_gr_2_05_redox	False	True	False	False	False
KWK_HMG_gr_3_05_redox	False	False	True	False	False
KWK_HMG_gr_4_05_redox	False	False	False	True	False
KWK_HMG_gr_5_05_redox	False	False	False	False	True

No significant pairs are found for HMG_5

HMG_60

	KWK_HMG_gr_1_60_redox	KWK_HMG_gr_2_60_redox	KWK_HMG_gr_3_60_redox	KWK_HMG_gr_4_60_redox	KWK_HMG_gr_5_60_redox
KWK_HMG_gr_1_60_redox	True	False	False	False	False
KWK_HMG_gr_2_60_redox	False	True	False	False	False
KWK_HMG_gr_3_60_redox	False	False	True	False	False
KWK_HMG_gr_4_60_redox	False	False	False	True	True
KWK_HMG_gr_5_60_redox	False	False	False	True	True

No significant pairs are found for HMG_60

LM_30

	KWK_LM_nat_1_30_redox	KWK_LM_nat_2_30_redox	KWK_LM_nat_3_30_redox	KWK_LM_nat_4_30_redox	KWK_LM_nat_5_30_redox
KWK_LM_nat_1_30_redox	True	False	False	False	False
KWK_LM_nat_2_30_redox	False	True	False	False	False
KWK_LM_nat_3_30_redox	False	False	True	False	False
KWK_LM_nat_4_30_redox	False	False	False	True	False
KWK_LM_nat_5_30_redox	False	False	False	False	True

No significant pairs are found for LM_30

LM_60

	KWK_LM_nat_1_60_redox	KWK_LM_nat_2_60_redox	KWK_LM_nat_3_60_redox	KWK_LM_nat_4_60_redox	KWK_LM_nat_5_60_redox
KWK_LM_nat_1_60_redox	True	False	False	False	False
KWK_LM_nat_2_60_redox	False	True	False	False	False
KWK_LM_nat_3_60_redox	False	False	True	False	False
KWK_LM_nat_4_60_redox	False	False	False	True	False
KWK_LM_nat_5_60_redox	False	False	False	False	True

No significant pairs are found for LM_60

PZ_30

	KWK_PZ_nat_1_30_redox	KWK_PZ_nat_2_30_redox	KWK_PZ_nat_3_30_redox	KWK_PZ_nat_4_30_redox	KWK_PZ_nat_5_30_redox
KWK_PZ_nat_1_30_redox	True	False	False	False	False
KWK_PZ_nat_2_30_redox	False	True	False	False	False
KWK_PZ_nat_3_30_redox	False	False	True	False	True
KWK_PZ_nat_4_30_redox	False	False	False	True	False
KWK_PZ_nat_5_30_redox	False	False	True	False	True

KWK_PZ_nat_5_30_redox and KWK_PZ_nat_3_30_redox are a pair at PZ_30

Results for winter

LM_60

	KWK_LM_nat_1_60_redox	KWK_LM_nat_2_60_redox	KWK_LM_nat_3_60_redox	KWK_LM_nat_4_60_redox	KWK_LM_nat_5_60_redox
KWK_LM_nat_1_60_redox	True	False	False	False	False
KWK_LM_nat_2_60_redox	False	True	False	False	False
KWK_LM_nat_3_60_redox	False	False	True	True	False
KWK_LM_nat_4_60_redox	False	False	True	True	False
KWK_LM_nat_5_60_redox	False	False	False	False	True

KWK_LM_nat_4_60_redox and KWK_LM_nat_3_60_redox are a pair at LM_60

LM_5

	KWK_LM_nat_1_05_redox	KWK_LM_nat_2_05_redox	KWK_LM_nat_3_05_redox	KWK_LM_nat_4_05_redox	KWK_LM_nat_5_05_redox
KWK_LM_nat_1_05_redox	True	False	False	False	False
KWK_LM_nat_2_05_redox	False	True	False	False	False
KWK_LM_nat_3_05_redox	False	False	True	False	False
KWK_LM_nat_4_05_redox	False	False	False	True	False
KWK_LM_nat_5_05_redox	False	False	False	False	True

No significant pairs are found for LM_5

HMG_60

	KWK_HMG_gr_1_60_redox	KWK_HMG_gr_2_60_redox	KWK_HMG_gr_3_60_redox	KWK_HMG_gr_4_60_redox	KWK_HMG_gr_5_60_redox
KWK_HMG_gr_1_60_redox	True	False	False	False	False
KWK_HMG_gr_2_60_redox	False	True	False	False	False
KWK_HMG_gr_3_60_redox	False	False	True	False	False
KWK_HMG_gr_4_60_redox	False	False	False	True	False
KWK_HMG_gr_5_60_redox	False	False	False	False	True

No significant pairs are found for HMG_60

HMG_30

	KWK_HMG_gr_1_30_redox	KWK_HMG_gr_2_30_redox	KWK_HMG_gr_3_30_redox	KWK_HMG_gr_4_30_redox	KWK_HMG_gr_5_30_redox
KWK_HMG_gr_1_30_redox	True	False	False	False	False
KWK_HMG_gr_2_30_redox	False	True	False	False	False
KWK_HMG_gr_3_30_redox	False	False	True	False	False
KWK_HMG_gr_4_30_redox	False	False	False	True	False
KWK_HMG_gr_5_30_redox	False	False	False	False	True

No significant pairs are found for HMG_30

HMG_10

	KWK_HMG_gr_1_10_redox	KWK_HMG_gr_2_10_redox	KWK_HMG_gr_3_10_redox	KWK_HMG_gr_4_10_redox	KWK_HMG_gr_5_10_redox
KWK_HMG_gr_1_10_redox	True	False	False	False	False
KWK_HMG_gr_2_10_redox	False	True	False	False	False
KWK_HMG_gr_3_10_redox	False	False	True	False	False
KWK_HMG_gr_4_10_redox	False	False	False	True	False
KWK_HMG_gr_5_10_redox	False	False	False	False	True

No significant pairs are found for HMG_10

HMG_5

	KWK_HMG_gr_1_05_redox	KWK_HMG_gr_2_05_redox	KWK_HMG_gr_3_05_redox	KWK_HMG_gr_4_05_redox	KWK_HMG_gr_5_05_redox
KWK_HMG_gr_1_05_redox	True	False	False	False	False
KWK_HMG_gr_2_05_redox	False	True	False	False	False
KWK_HMG_gr_3_05_redox	False	False	True	False	False
KWK_HMG_gr_4_05_redox	False	False	False	True	False
KWK_HMG_gr_5_05_redox	False	False	False	False	True

No significant pairs are found for HMG_5

HM_5

	KWK_HM_nat_1_05_redox	KWK_HM_nat_2_05_redox	KWK_HM_nat_3_05_redox	KWK_HM_nat_4_05_redox	KWK_HM_nat_5_05_redox
KWK_HM_nat_1_05_redox	True	False	False	False	False
KWK_HM_nat_2_05_redox	False	True	False	False	False
KWK_HM_nat_3_05_redox	False	False	True	False	False
KWK_HM_nat_4_05_redox	False	False	False	True	False
KWK_HM_nat_5_05_redox	False	False	False	False	True

No significant pairs are found for HM_5